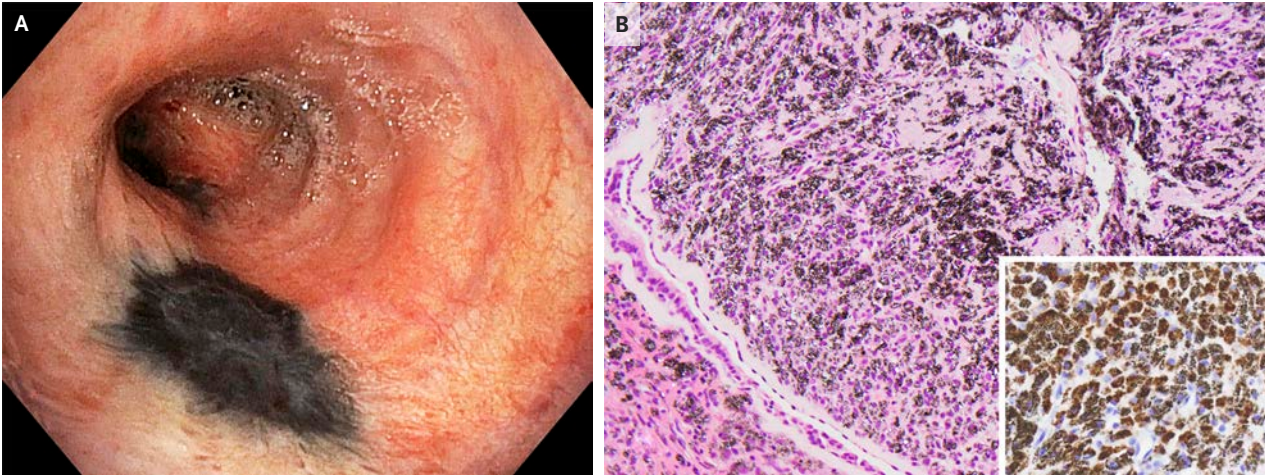


## IMAGES IN CLINICAL MEDICINE

## Bronchial Anthracosis



Fredrik Eika, M.D.<sup>1</sup>

Torpiki Abdul Abdulhaq, M.D.<sup>2</sup>

<sup>1</sup>Diakonhjemmet Hospital, Oslo, Norway;

<sup>2</sup>Oslo University Hospital, Oslo, Norway.

Fredrik Eika can be contacted at  
eikafredrik@gmail.com.

A 77-YEAR-OLD MAN UNDERGOING BRONCHOSCOPY FOR EVALUATION OF suspected lung cancer was noted to have black patches on the bronchial mucosa. He had a smoking history of 50 pack-years and a history of pulmonary tuberculosis treated with left upper lobectomy. During bronchoscopy, areas of black discoloration were seen on the mucosa of the left main bronchus and left segmental bronchi (Panel A). A biopsy sample showed ciliated respiratory epithelium with subepithelial accumulation of macrophages containing black pigment (Panel B, hematoxylin and eosin stain). Also noted was the deposition of small, birefringent dust particles of unknown origin, which were thought to possibly be silicates. On immunohistochemical staining, abundant pigment-laden macrophages were identified by their diffuse cytoplasmic positivity for CD68 (Panel B, inset). A diagnosis of bronchial anthracosis — a benign, often incidental finding indicating carbon pigment deposition in the mucosa — was made. Bronchial anthracosis is associated with prolonged exposure to biomass smoke or environmental dust. On further history taking, the patient reported having socialized around open fires for decades of his life in the Middle East, as well as having been exposed to substantial amounts of dust while incarcerated for 10 years. No treatment was recommended for the bronchial anthracosis; subsequent management and follow-up focused on the lung cancer that had been confirmed by computed tomography (CT) and CT-guided biopsy.

DOI: 10.1056/NEJMicm2516602

Copyright © 2026 Massachusetts Medical Society.